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Intro to Probability, Kolmogorov Complexity

Computer Science Mentors 70

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Introduction to Kolmogorov Complexity

1. Recall from HW 7 that the *Kolmogorov complexity* of a string $K(x)$ is the length of an optimally-compressed copy of x; that is, $K(x)$ is the length of the shortest program that returns x. Let's explore this definition a little bit.

   (a) True or false? The Kolmogorov complexity of any string is always smaller than the length of the string itself.

   (b) True or false? When you use a program to "zip" files on your computer into a compressed archive, the resulting .zip file has a smaller Kolmogorov complexity than the original files.

   (c) In plain English, write the shortest possible representation of the following strings. There are multiple possible solutions because of the constraints of the English language.

      (a) `cs70cs70cs70cs70cs70cs70cs70`

      (b) `3.14159265358979323846264...`

      (c) `We're no strangers to love / You know the rules and so do I / A full commitment's what I'm thinking of / You wouldn't get this from any other guy / I just wanna tell you how I'm feeling / Gotta make you understand / Never gonna give you up / Never gonna let you down / Never gonna run around and desert you ...`

Introduction to Probability

1. Suppose two integers a and b are drawn uniformly from $[-n .. n]$, that is $a, b \in \mathbb{Z}$ and $-n \leq a, b \leq n$.

   (a) Define a probability space for (a, b). Does each sample point occur with uniform probability?

(b) Find the probability that $\max\{0, a\} = \min\{0, b\}$.

(c) Find the probability that $|a - b| \leq k$. You may assume $k < \frac{n}{2}$.

(d) Suppose we choose two closed intervals $u = [a \mathinner{..} b], v = [c \mathinner{..} d]$ uniformly at random from $[-n \mathinner{..} n]$. What is the probability that u is enveloped by v, meaning that $u \subset v$ and $c < a \leq b < d$. What happens to this probability as n approaches ∞?

2. Alex and Shruti are playing Yahtzee, a game involving rolling 5 dice.

   (a) First, define a probability space representing the possible outcome of Alex or Shruti's rolls of the 5 dice. Assume all dice are fair and labeled 1 through 6.

Alex and Shruti each roll 1 die to see who goes first. The person with the higher roll goes first, and in case of a tie, they both roll their die again.

(b) What's the chance Shruti rolls a higher number on the first roll?

(c) What's the chance Shruti goes first?

(d) They finally begin playing. Partway through the game, Alex is missing the "three of a kind" category while Shruti is missing the "four of a kind" category. What is the probability of rolling...

1. exactly 3 of a kind?

2. exactly 4 of a kind?

3. Which one is more likely? 3 of a kind or 4 of a kind?

Inclusion-Exclusion Principle, Bayes' Theorem

1. Tri-State Area is experiencing bad weather because of Doctor Doofenshmirtz "Gloomy - inator". It is always at least rainy, cloudy or windy, but because the inator is random we don't exactly know what it would be like. It rains with 0.5 probability, gets windy with 0.65 probability and gets cloudy with 0.45 probability. We experience at least 2 of these together with probability 0.45. Help Agent P find the probability that all 3 of these happen together.

2. You own a pizzeria. You observe that one of your customers, Andy, buys a cheese pizza on Saturday with probability 0.3 and on Sunday with probability 0.6.

   (a) If Andy's pizza purchasing habits on Sunday is independent from his pizza purchasing habits on Saturday, what is the probability that he buys pizza on a given weekend?

   (b) If Andy buying a pizza on Saturday means that he will not buy pizza on Sunday, what is the probability that he buys pizza on a given weekend (i.e if he buys pizza on one day, he is guaranteed to not buy a pizza the next day)? Note that the probability that he buys a pizza on Sunday, 0.6, is *not* a conditional probability, i.e. it is not conditioned on whether he buys a pizza on Saturday.

   (c) Suppose we don't know how Andy's pizza purchasing habits on Sundays depends on whether he bought a pizza on the preceding Saturday. Given that Andy buys pizza on a given weekend with probability 0.65, what is the probability that he buys pizza both days?

3. (*Note: before students attempt this problem, it is highly recommended that they attempt problem 1d*)

   Suppose Ezekiel places m rectangles on an $n x n$ grid, each chosen uniformly at random and independent from all others. Lower bound the probability that no rectangle is enveloped by another rectangle (that is, without any edges touching, the area of one rectangle is located entirely inside the area of another rectangle). Assuming n is sufficiently large, express this bound as a function of m